Topology-Aware Mapping for Photonic Interconnects in TPU-Class LLM Accelerators

6.5930 Project 12: Evaluate a New Technology

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Hardware Architectures for Deep Learning

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Emerging technology: Lightmatter Passage photonic interconnects

One-Slide Thesis

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Best modeled inference speedup: Gemma 4 26B MoE on a 9216-chip Ironwood-class pod.

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Inner AccelForge mapping signatures changed under photonic TPU-cube variants.

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Distributed schedules changed when multicast and pooled-HBM became cheap.

Claim

Photonic interconnect is not just a lower pJ/bit link. It changes the useful distributed schedule above the local mapping: multicast cache fill, remote pooled-HBM streaming, and flatter topologies become rational.

Speaker note. The short version is: the local GEMM dataflow stays stable, but the system-level schedule changes. That is the clean way to reconcile the mapping results with the compiler/dataflow results.

Why This Is a TPU/LLM Problem

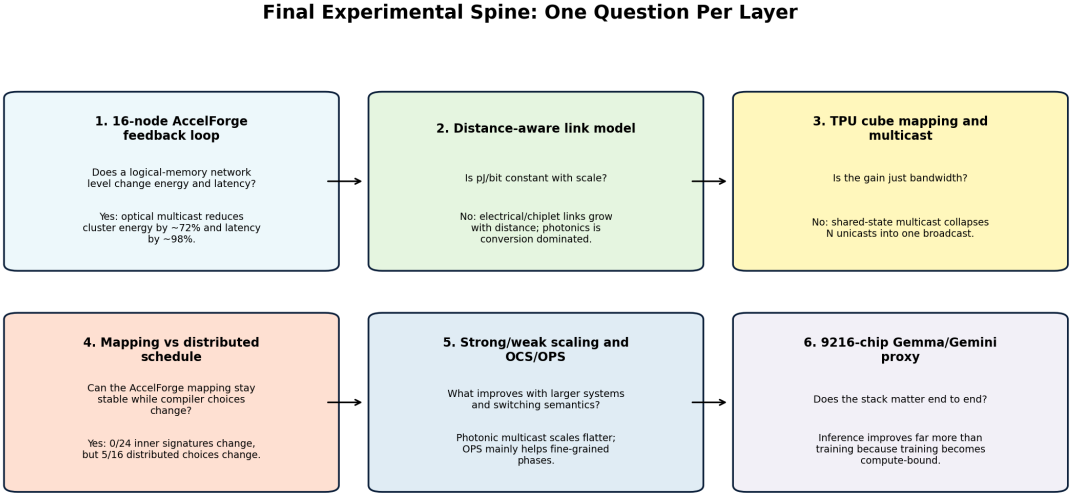
- TPU7x/Ironwood public specs: 9216 chips per pod, 192 GiB HBM/chip, 1200 GB/s bidirectional ICI/chip.
- LLM stages move activations, KV state, expert tokens, gradients, and weights across chips.
- Google software already treats the system as a sharded device mesh: XLA/GSPMD inserts collectives from sharding annotations.

Project question

If tomorrow's TPU pod had Lightmatter-style photonic ICI, would we only reduce link cost, or would we choose different memory placements and distributed schedules?

Speaker note. We are not trying to model Google's confidential Gemini internals. We use public Gemma 4 anchors and explicit Gemini-class stress proxies to stress communication patterns.

The Final Experiment Spine



Class Concept Map: What "Mapping" Means Here

1. **Einsum/workload:** `single_matmul.yaml`, $O_{m,n} = \sum_k A_{m,k} W_{k,n}$.
2. **Architecture:** logical cluster memory above per-chip HBM, UB, registers, and MACs.
3. **AccelForge mapping:** storage, temporal, spatial, and compute nodes; this is local dataflow, tiling, and dataplacement.
4. **Distributed schedule:** XLA/GSPMD-like choice among collectives, remote pooled HBM, replicated cache, and photonic multicast cache fill.

Important distinction

The inner mapping can stay the same while the distributed schedule changes. In this project, that is exactly what happens.

Why AccelForge still matters

It supplies the action counts and logical-memory traffic that the topology model prices.

Speaker note. This slide addresses the main vocabulary issue. We reserve "dataflow" for the local loop/order choice and say "distributed schedule" for the compiler-visible pod-level choice.

Network Model Fed Back Into AccelForge

$$E_{\text{net}} = B(e_c + h\,\ell(N)e_{tx})$$

$$T_{\text{net}} = \frac{B}{BW/C(N)} + ht_r + h\,\ell(N)t_p + t_s$$

- B: bits from mapper action counts.
- h: topology hop count.
- $\ell(N)$: average path distance.
- $C(N)$: contention.
- t_s: OCS reconfiguration / OPS packet term.

Fixed-point loop

AccelForge mapping → logical-memory traffic → topology energy/latency → updated `LogicalClusterMemory` → rerun.

Multicast rule

Electrical/chiplet pay repeated unicasts. Multicast-capable photonics pays the unique payload once.

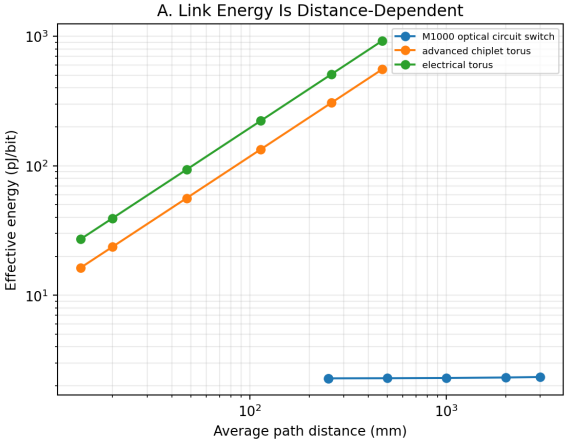
Speaker note. This is the modeling contract. The project is not just a post-hoc link calculator; the logical memory level receives the network cost and the mapper is rerun.

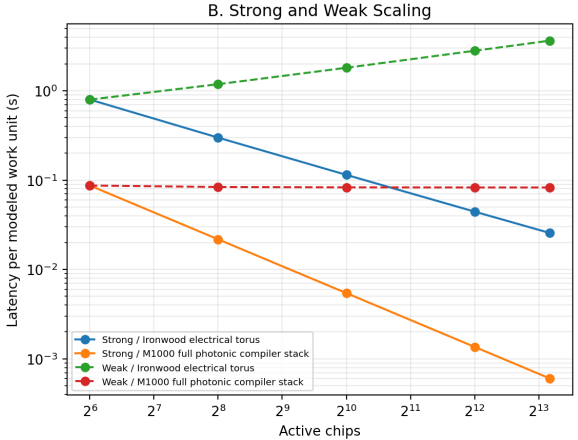
Assumptions Kept Explicit

Item	Value	Role
Ironwood pod	9216 chips; 192 GiB HBM/chip; 1200 GB/s ICI/chip	Full-system baseline
M1000	57.3 Tb/s one-way; 2.3 pJ/bit incl. laser	Main photonic fabric
EVK100	3.2 Tb/s class; 1.9–3.2 pJ/bit	Conservative and optimistic sensitivity
Electrical	0.2 pJ/bit conversion + 1.96 pJ/bit/mm	Distance-sensitive baseline
Chiplet	3.5 Tb/s; 6 pJ/bit per hop	Strong non-optical comparison
Compiler layer	collectives, remote HBM, replicated cache, photonic multicast	XLA/GSPMD analogy

Speaker note. Everything that moves a headline number is visible here. If someone asks what was fixed, the answer is compute, MAC/register/UB/HBM, and HBM cost. We vary interconnect, topology, and distributed schedule.

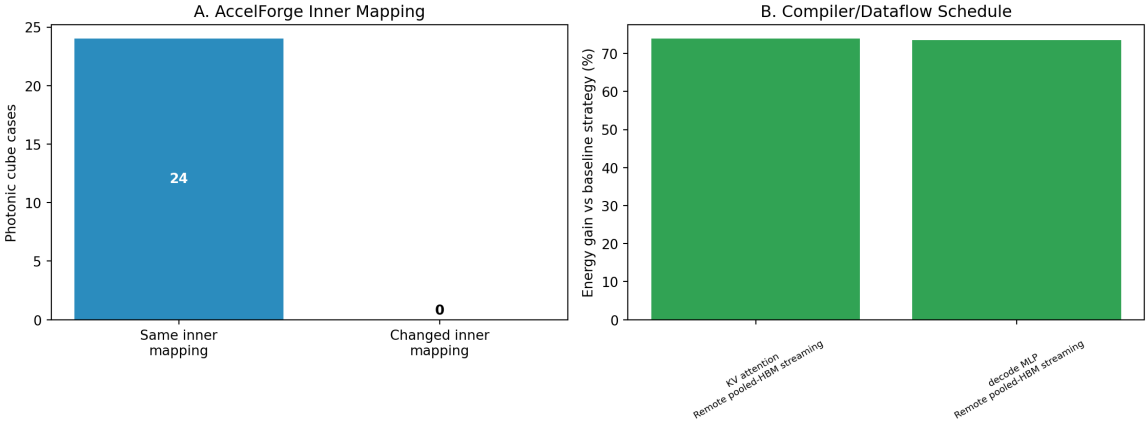
Result 1: Distance Changes the Scaling Law





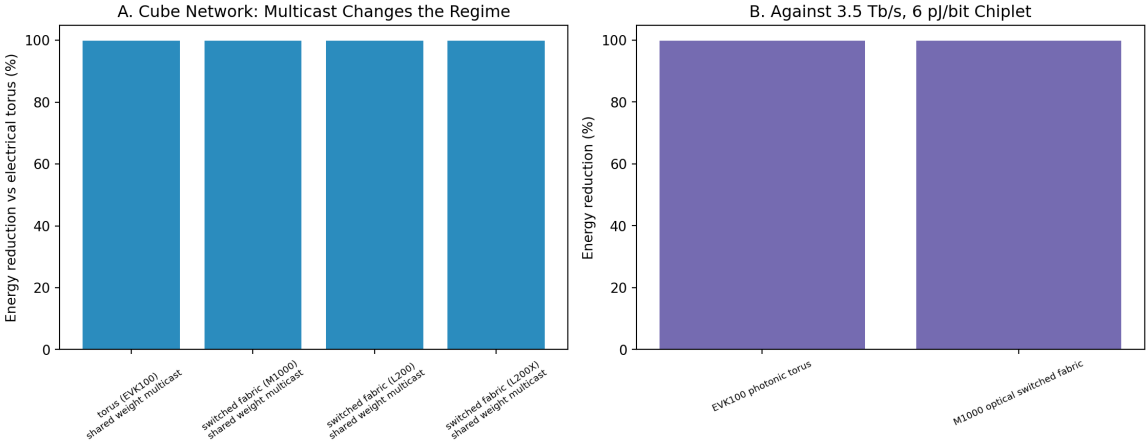
Speaker note. This is why constant pJ/bit was too simple. Electrical is good at very short reach, but its transmission term grows with distance. Photonics pays conversion and then stays nearly flat. Strong scaling and weak scaling are shown separately so we do not confuse lower latency on fixed work with higher aggregate throughput.

Result 2: Local Mapping Stable, Distributed Schedule Changes



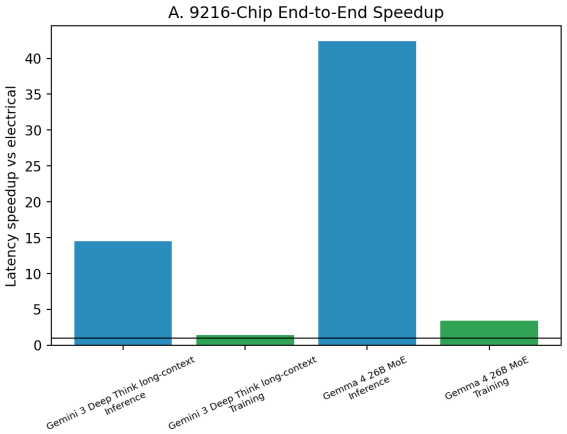
Speaker note. The left panel is the AccelForge result: the local mapping does not change under the photonic swap. The right panel is the compiler-visible result: remote HBM and multicast policies become better. This is the central resolution of the apparent contradiction.

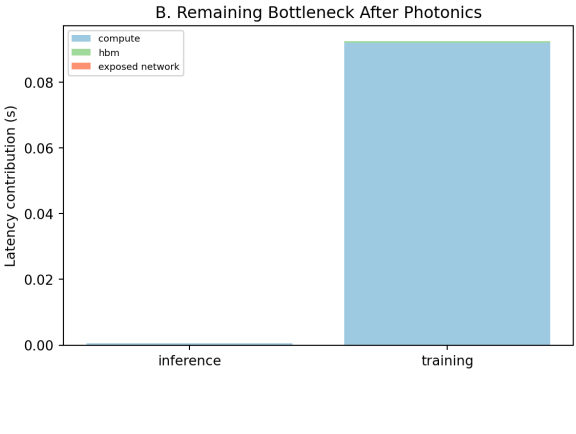
Result 3: Multicast Beats a Strong Chiplet Baseline



Speaker note. The chiplet baseline is deliberately not weak: 3.5 Tb/s and 6 pJ/bit/hop. Photonics still wins on shared state because the communication pattern changes from repeated unicast to one logical broadcast. This result would disappear for purely local traffic, which is why the paper does not claim a universal speedup.

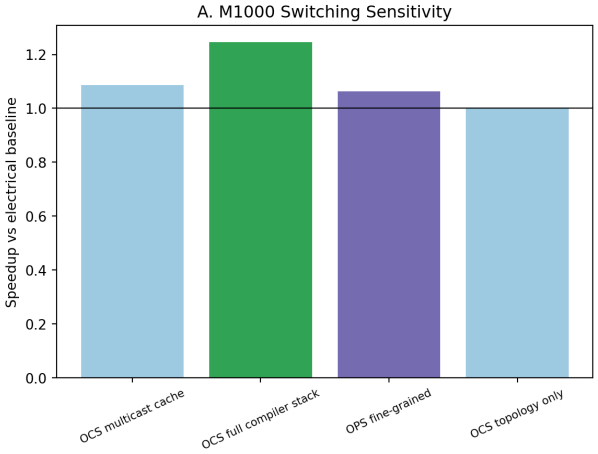
Result 4: End-to-End Superpod Impact





Speaker note. Inference gets the large gain because decode, KV/cache traffic, and expert routing expose network stalls. Training improves too, but once collective traffic shrinks the remaining work is mostly compute. That is a useful limitation, not a weakness of the study.

Result 5: Switching Mode and Model Scope



B. Scope: what the model does and does not claim

Included

- AccelForge actions
- HBM
- distance energy
- hop contention
- multicast
- stage overlap

Excluded / future work

- packet queues
- thermal drift
- laser aging
- XLA codegen
- area cost
- model accuracy

Speaker note. OCS is strong when traffic has large phases that can amortize setup. OPS is a sensitivity point for fine-grained phases. The right panel is intentionally blunt: it says exactly what the model includes and what it does not include.

Headline Results

Layer	Result	Interpretation
16-node AccelForge	71.95% energy, 97.57% latency reduction	Logical network cost matters
TPU cube network	99.93% energy reduction	Multicast collapses repeated traffic
TPU cube mapping	0/24 inner mappings changed	Local dataflow is stable
Distributed schedule	5/16 choices changed	Compiler-visible mapping changes
Compiler strategy	73.86% best energy gain	Remote HBM/multicast can be optimal
Gemma 4 MoE inference	42.43× speedup, 94.05% energy reduction	Decode/cache/expert routing benefits most
Gemma 4 MoE training	3.37× speedup	Communication improves, then compute dominates

Speaker note. This is the table to memorize for Q&A. It has a positive result, a negative result, a compiler result, and the end-to-end result.

What Worked, What Did Not

Worked

- Distance-aware photonic links at pod scale.
- Hardware multicast for shared state.
- Remote pooled-HBM streaming and multicast cache fill.
- Decode-heavy and MoE inference.

Limited

- Local GEMM mappings did not change in the tested mapspace.
- Node-to-node unicast gains were modest.
- Training became compute-bound after collectives shrank.
- Packet-level queues, area/cost, thermal drift, and exact XLA codegen are future work.

Speaker note. This slide is there so the claims are qualified. The project is strongest where traffic is shared, broadcast, or remote-cache-like; it is weaker where traffic is local or compute-bound.

Reproducibility

Commands

```
make study
make notebook
make test
make deliverables
```

- Raw data: `results/*.csv`.
- Final submission figures/tables: `results/presentation/`.
- AccelForge architecture: `arch/tpu_lightmatter_cluster.yaml`.
- Workload template: `workload/single_matmul.yaml`.
- Regression suite checks technology constants, distance crossover, multicast behavior, mapping invariance, strong/weak scaling, and OCS/OPS.

Speaker note. This is the answer to "how do I reproduce the figures?" The final presentation folder is small on purpose; the larger CSV directory is for auditability.

References


Google Cloud, “TPU7x (Ironwood),” https://docs.cloud.google.com/tpu/docs/tpu7x


Lightmatter, “Passage M1000 EVK,” https://lightmatter.co/products/passage-m1000-evk/


Lightmatter, “Passage EVK100,” https://lightmatter.co/products/passage-evk100/


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PyTorch/XLA, “SPMD User Guide,” https://docs.pytorch.org/xla/master/spmd.html


M. Gilbert, T. Andrulis, V. Sze, and J. Emer, “The Turbo-Charged Mapper,” arXiv:2602.15172, 2026.

Speaker note. The project uses official public specs where available and labels stress proxies where Google does not disclose model internals.

Takeaway

Final answer

Photonics matters most when the software and memory system can expose multicast, pooled HBM, and flatter topology to the schedule. The local accelerator mapping can remain the same; the pod-level data movement strategy changes.

Questions